



SIMATS ENGINEERING

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Student Names:

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Title: CNN-Based Traffic Sign Recognition System

Assignment Description:

This assignment focuses on designing and implementing a CNN-based Traffic Sign Recognition System covering the complete deep-learning workflow — data preparation, preprocessing, CNN model design, training, evaluation and analysis. We use a compact CNN architecture to extract important visual features from traffic-sign images while keeping the model suitable for practical training in Google Colab. The images are resized and normalized before training, while data augmentation using rotation, shifting and zooming is applied to improve generalization. A controlled subset of the GTSRB dataset is used for the experiment to maintain manageable training time while preserving the main traffic-sign classification problem. The trained model is evaluated using accuracy, precision, recall, F1-score and a confusion matrix, while sample predictions are used to examine classification confidence and errors. The main challenge is distinguishing visually similar traffic-sign categories, where small differences in shape, symbols or numbers can lead to misclassification. The project demonstrates how CNN architecture, preprocessing and augmentation choices influence recognition performance and computational cost. The work maps to **SDG 9 (Industry, Innovation and Infrastructure)** and **SDG 11 (Sustainable Cities and Communities)** through its potential application in intelligent transportation and driver-assistance systems

- **CO1** (learning algorithms, backpropagation) — attained through implementing and training the CNN using gradient-based optimization and cross-entropy loss.
- **CO2** (designing deep architectures) — attained through CNN architecture design, layer configuration and justification for traffic-sign image classification.
- **CO3** (developing and tuning CNNs for vision tasks) — attained through preprocessing, augmentation, model training, evaluation using accuracy/precision/recall/F1 and confusion-matrix analysis.

Guide Signature

Student Signature